



REPORT SOCIAL MEDIA & WEB ANALYTICS

CLICKBAIT AND EMOTIONAL FRAMING: AN ANALYSIS OF YOUTUBE NEWS CHANNELS

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January 30, 2026

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RESEARCH QUESTION & OBJECTIVES

How do alternative and mainstream news channels differ in their use of clickbait and emotional language on YouTube?

1

How do these strategies relate to audience engagement and interaction?

2

- Examine how news framing adapts to algorithmic visibility on YouTube
- Compare mainstream and alternative channels' use of clickbait and emotions
- Assess the relationship between emotional framing and audience engagement
- Identify differences in interaction patterns and discussion dynamics
- Understanding polarization in digital news environments



METHODOLOGY OVERVIEW



Data Collection & Cleaning

- **Tool:** YouTube Data API v3
- **4 channels:** BBC News, CNN, The Young Turks, The Jimmy Dore Show
- **Time window:** Oct 2025 – Jan 2026
- **Size:** 2,608 videos, 451,047 comments, 10456 replies

01



Linguistic & Emotional Analysis

- Clickbait indicators (lexical & structural)
- **NLP cleaning:** lemmatization, stopwords, emojis, URLs (SpaCy)
- Topic modeling (LDA)
- Sentiment analysis (VADER)
- Emotion analysis (NRClex)

02



Network Analysis

- *User interaction networks (replies)*
- **Replies** were **sampled** from the **top 1,500 most liked comments** per channel
- **Time window:** 5 to 20 January 2026
- **Metrics:** Nodes, Edges, Average degree, Density, Connected components, In-degree centrality, Out-degree centrality

03

METADATA REVIEW

- Two datasets

content_df:

- Comment-level granularity
- Shape (451047,10)

interaction_df:

- Reply-level granularity
- Shape (10456,17)

Field	Short Description
channel	Name or ID of the YouTube channel that published the video.
video_id	Unique identifier of the YouTube video.
video timestamp	Date and time when the video was published.
title	Title of the video.
description	Text description of the video.
comment id	Unique identifier of the comment.
comment author	Name or ID of the comment's author.
comment text	Text content of the comment.
comment likes	Number of likes received by the comment.
comment timestamp	Date and time when the comment was posted.

Field	Short Description
parent_id	ID of the main comment to which the reply refers.
reply id	Unique identifier of the reply.
reply author	Name or ID of the reply's author.
reply text	Text content of the reply.
reply likes	Number of likes received by the reply.
reply timestamp	Date and time when the reply was posted.

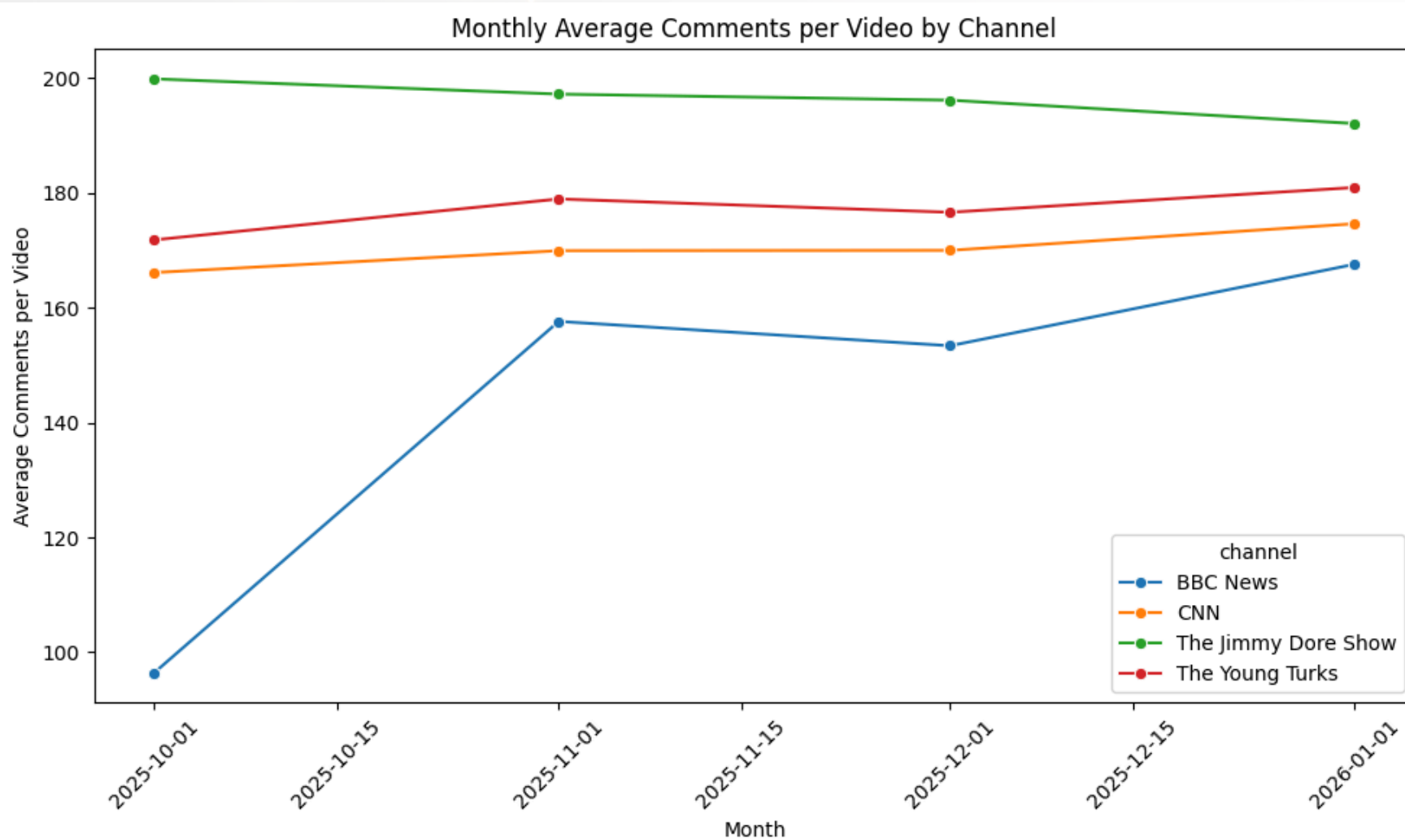
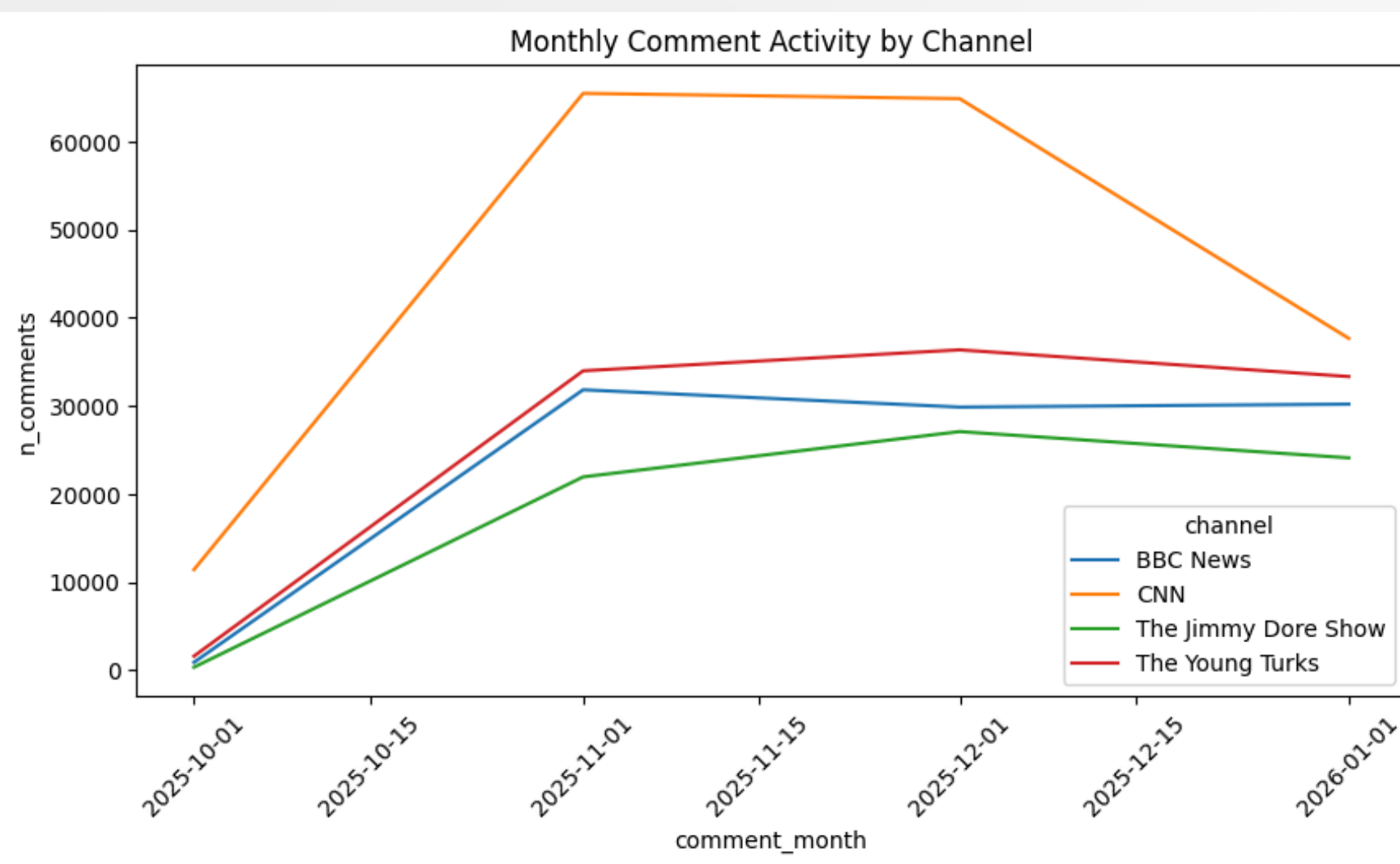
KEY INSIGHTS OVERVIEW

Volume vs Intensity

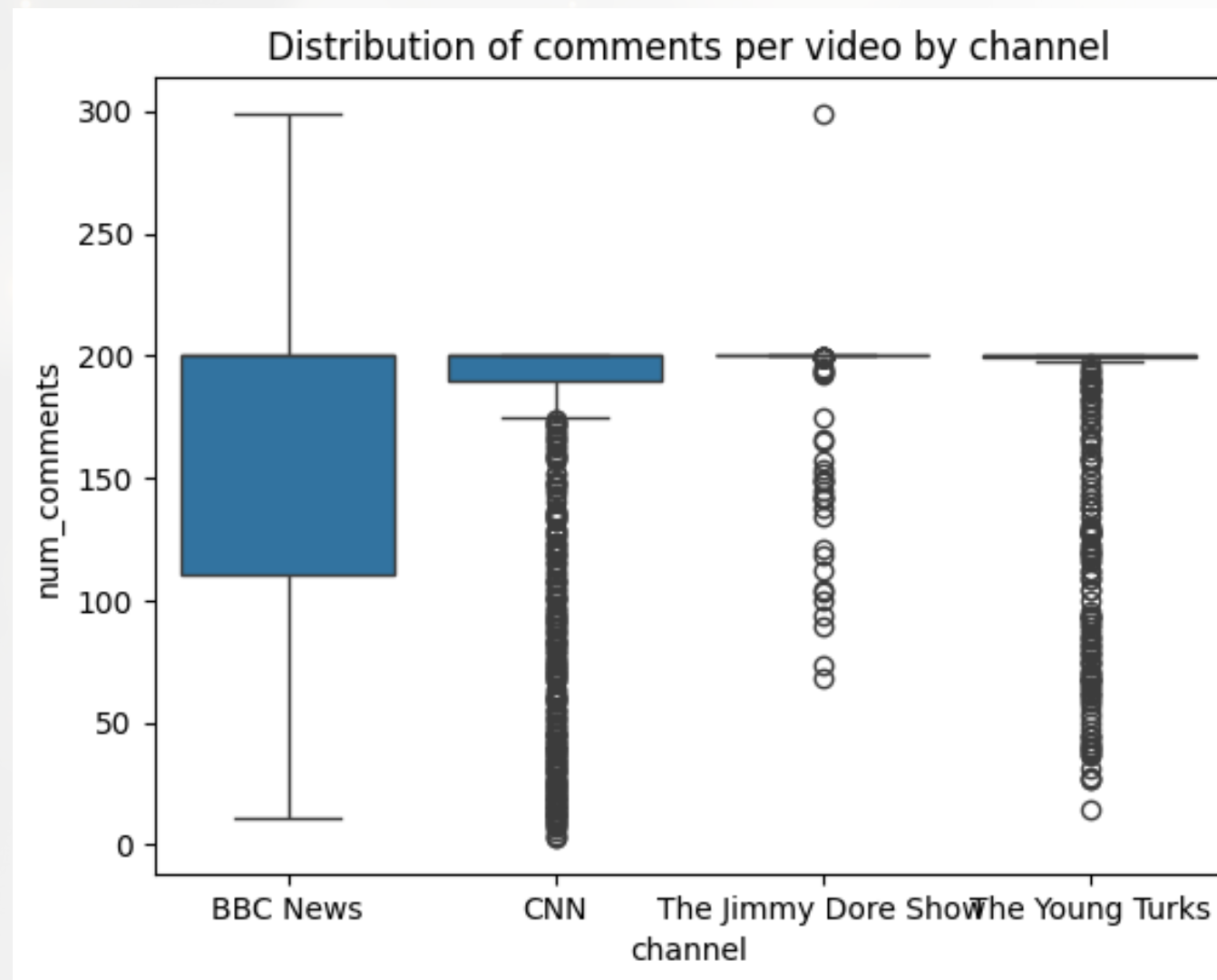
- More videos published → higher total comment volume
- CNN and The Young Turks lead with 1,053 and 590 unique videos → expected highest share of total comments

Comment volume share (of total across channels):

- CNN: 39.8%
 - The Young Turks: 23.3%
 - BBC News: 20.6%
 - The Jimmy Dore Show: 16.3%
-
- **Alternative** channels show a **consistent commenting** base across nearly every video
 - No contradiction: volume reflects channel size, intensity (comments/video) reflects audience loyalty
 - Notably, **CNN mirrors the high & stable per-video engagement** of alternative channels → explored in next slide



KEY INSIGHTS OVERVIEW



- **BBC News** → highly variable engagement across videos (some very low, many moderate, and a few highly commented) → **heterogeneous engagement** pattern, likely **content-driven**
- **CNN, The Jimmy Dore Show, and The Young Turks** → most videos reach similarly **high engagement** levels (~180–200 comments) with limited variability → presence of lots of **poorly commented videos** → **polarized/hit-driven engagement with loyal community**

KEY INSIGHTS OVERVIEW

Figure: Example of clickbait lexical indicator.

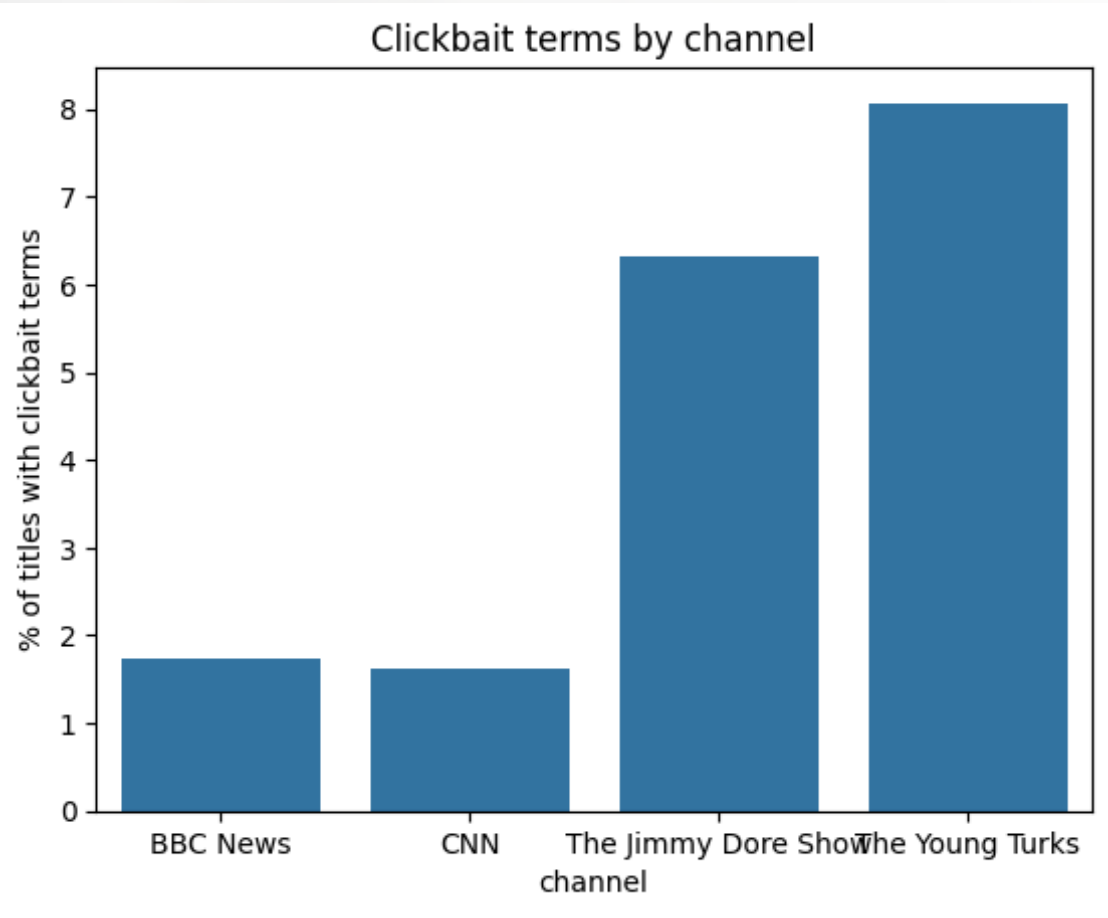


Figure: Correlation Matrix between Clickbait indicators and engagement; Weak correlation;

	punctuation_score	title_length	uppercase_ratio	num_comments
punctuation_score	1.000000	-0.047111	0.380170	0.014946
title_length	-0.047111	1.000000	-0.366325	-0.100358
uppercase_ratio	0.380170	-0.366325	1.000000	0.108333
num_comments	0.014946	-0.100358	0.108333	1.000000

Clickbait indicators vs Engagement:

- Alternative channels display stronger presence on the usage of clickbait structures language:
 - higher % of clickbait terms
 - higher average punctuation score (presence of [!?.])
 - lower average title length
 - higher uppercase ratio in titles
- **No relevant correlation** between **clickbait** indicators chosen and **engagement** (number of comments in video)
- Engagement depends on:
 - topic of the video
 - politic event
 - channel notoriety...

KEY INSIGHTS OVERVIEW

Linguistic & Topic Analysis:

- **Mainstream (BBC/CNN):** "cooler" and more branded framing, with emphasis on Trump as an event/news item. Fewer emotional words. (countries, presidents' names, "kill", "shooting")
- **Alternative (TYT/Jimmy Dore):** much more emotional and polarizing framing, with words that evoke anger/indignation ("expose", "lie", "attack", "assassination"). Israel emerges as a key word.
- Topic modeling confirms: **alternative titles** are more "**personal**" (host/conservative names) and **polemical** → strategy to generate emotional engagement and comments → the model identifies separate topics
- Mainstream channels treat mainly geopolitics, wars and scandals → all topics very similar to each other (uniform)

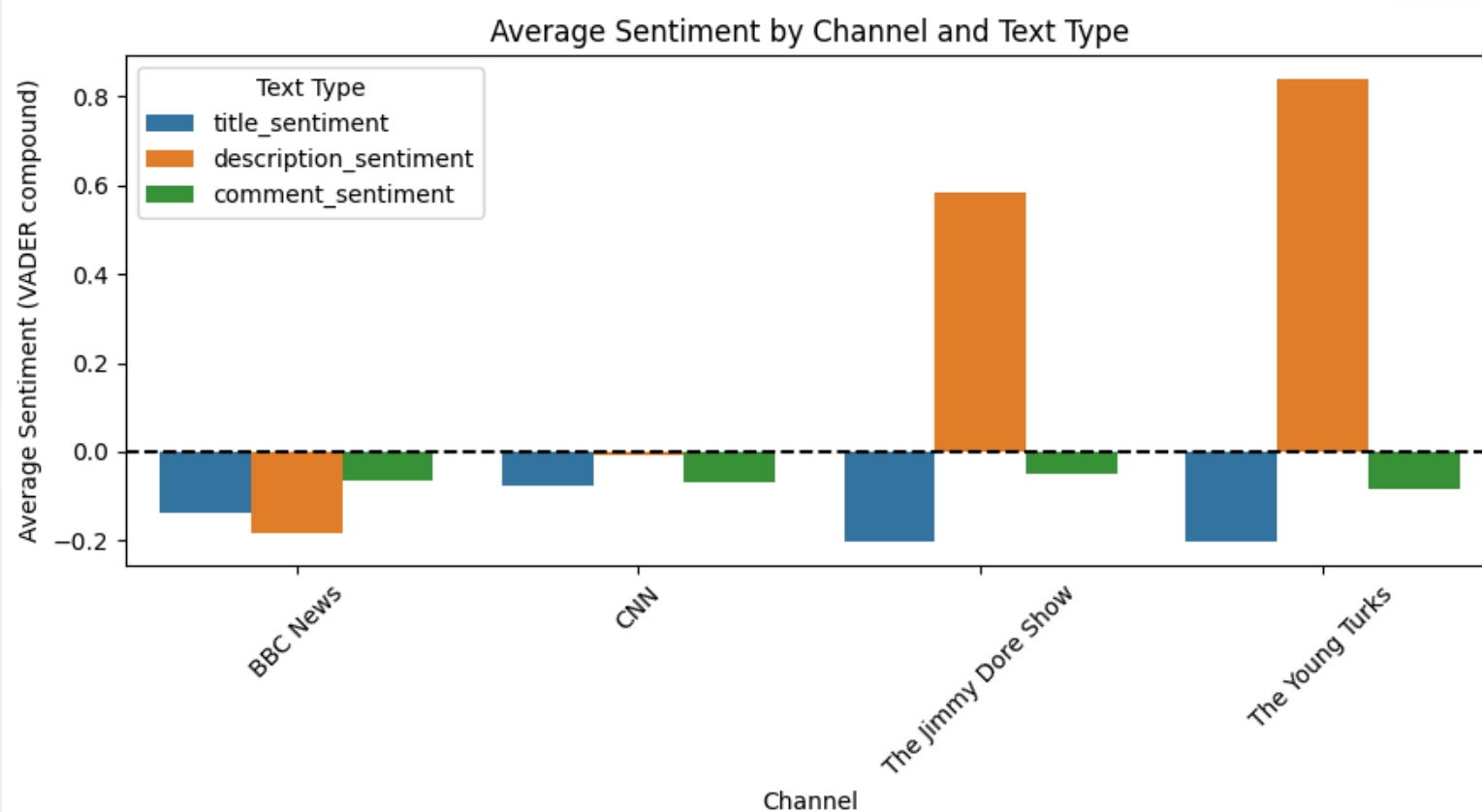
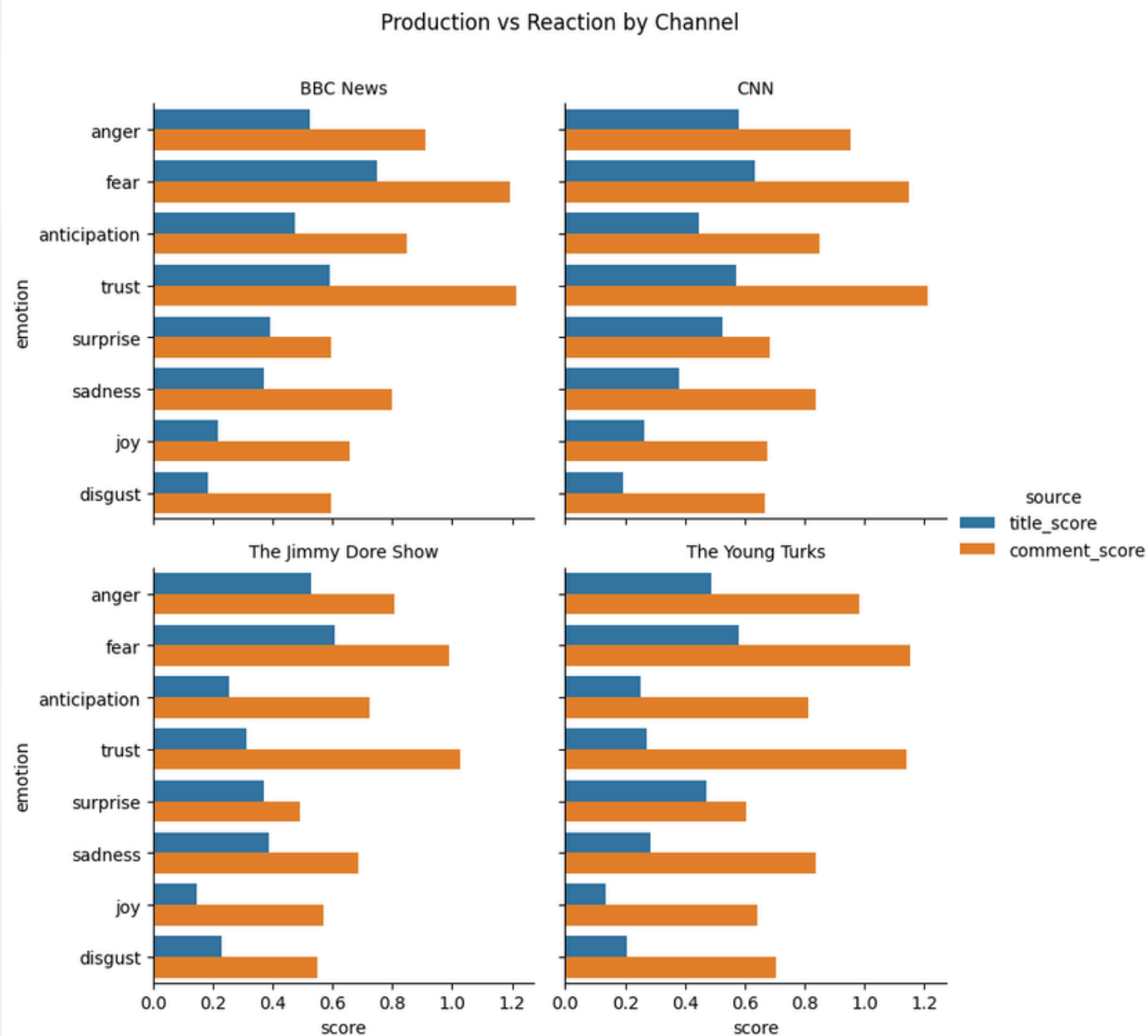
Example of topics identified in titles:

For The Young Turks:

- *Topic 0: Trump + Epstein + Israel ("get", "report", "shapiro", "levin") → Scandals + criticism of the right.*
- *Topic 1: Trump + files/news ("mamdani", "netanyahu", "drop") → Breaking news/Scandals + domestic/foreign policy.*
- *Topic 2: Israel + war ("iran", "expose", "attack", "war") → Criticism of Israel/U.S. policy.*
- *Topic 3: Host + debate ("cenk", "ana", "ben", "shapiro", "maga", "gaza") → Ideological debate, anti-MAGA.*
- *Topic 4: Trump + Tucker + Israel ("carlson", "ice", "greene") → Criticism of conservatives + immigration.*

Note: Linguistic analysis shows tendency of monetization and community building in alternatives' channels descriptions ("twitter", "subscribe", channels' name, "livestream", "premium"); its consequences are evident in sentiment analysis

KEY INSIGHTS OVERVIEW



Sentiment & Emotion in Titles and Descriptions

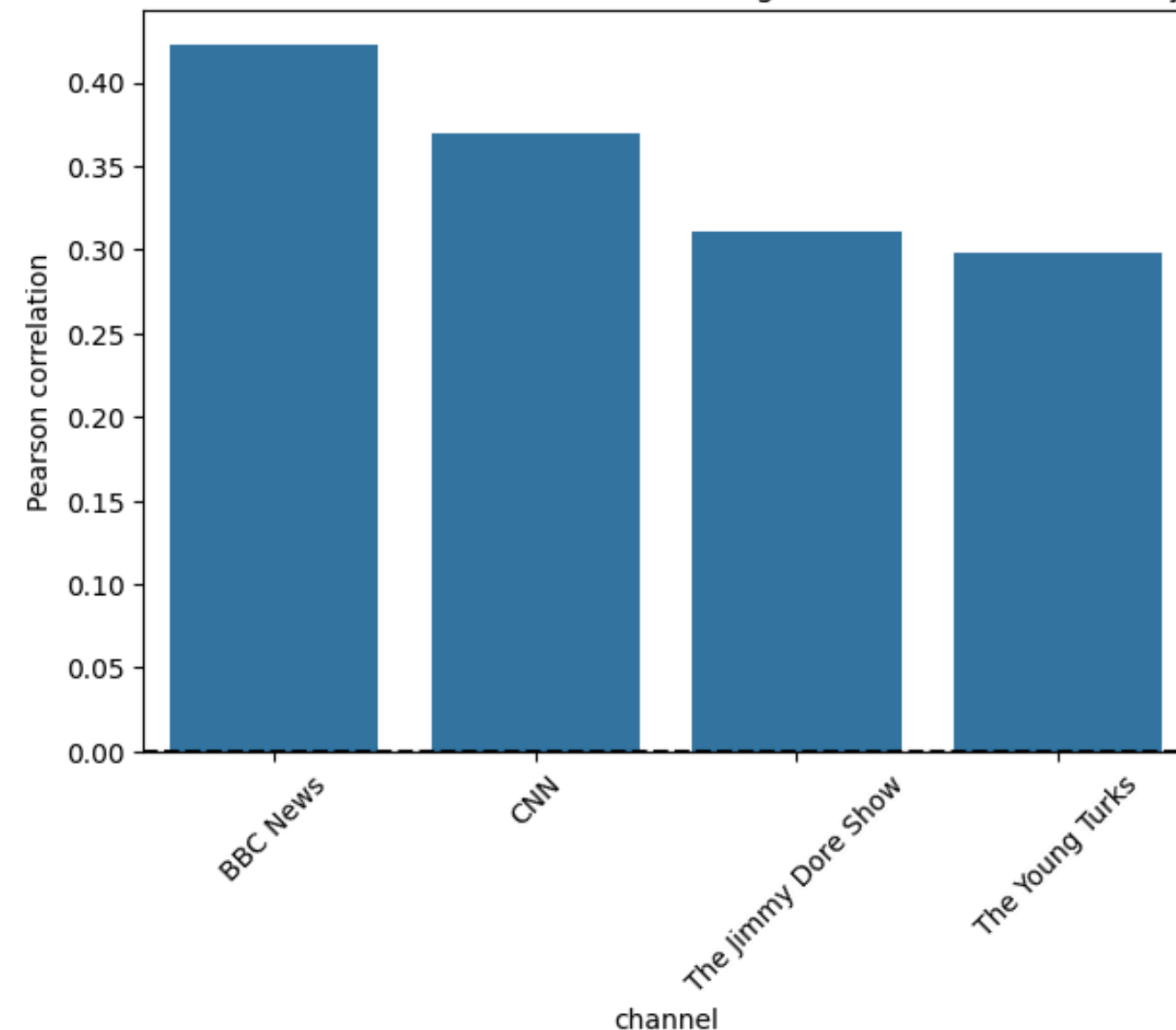
- **Fear and anger dominate titles across all channels:** emotional framing is central to news on YouTube
- **Mainstream** channels (BBC/CNN): **consistent emotional tone** between titles and descriptions
- **Alternative channels** (TYT/Jimmy Dore): strong emotional split
 - Titles: high fear, anger, surprise
 - Descriptions: dominated by trust, anticipation, joy: branding & audience retention strategy

User Comments

- Comments show higher emotional intensity than titles (all channels)
- Dominant emotions: trust + fear and anger (paradoxical)
- Bimodal sentiment in comments: peaks near neutral + moderate negative values (polarized reactions)
- Emotional framing
 - provokes hostility
 - Strengthens perceived legitimacy and group belonging

KEY INSIGHTS OVERVIEW

Correlation between Title Sentiment and Average Comment Sentiment by Channel



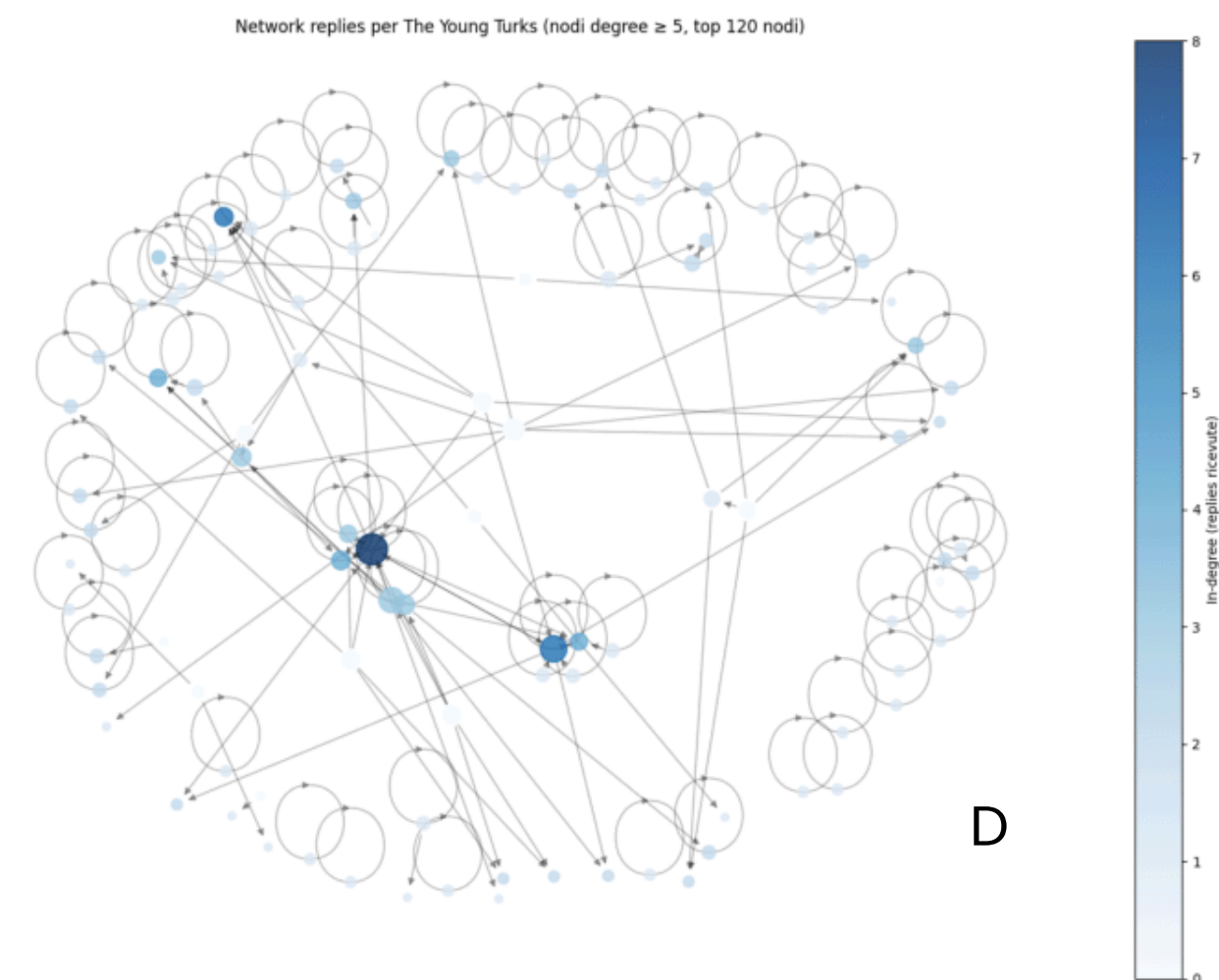
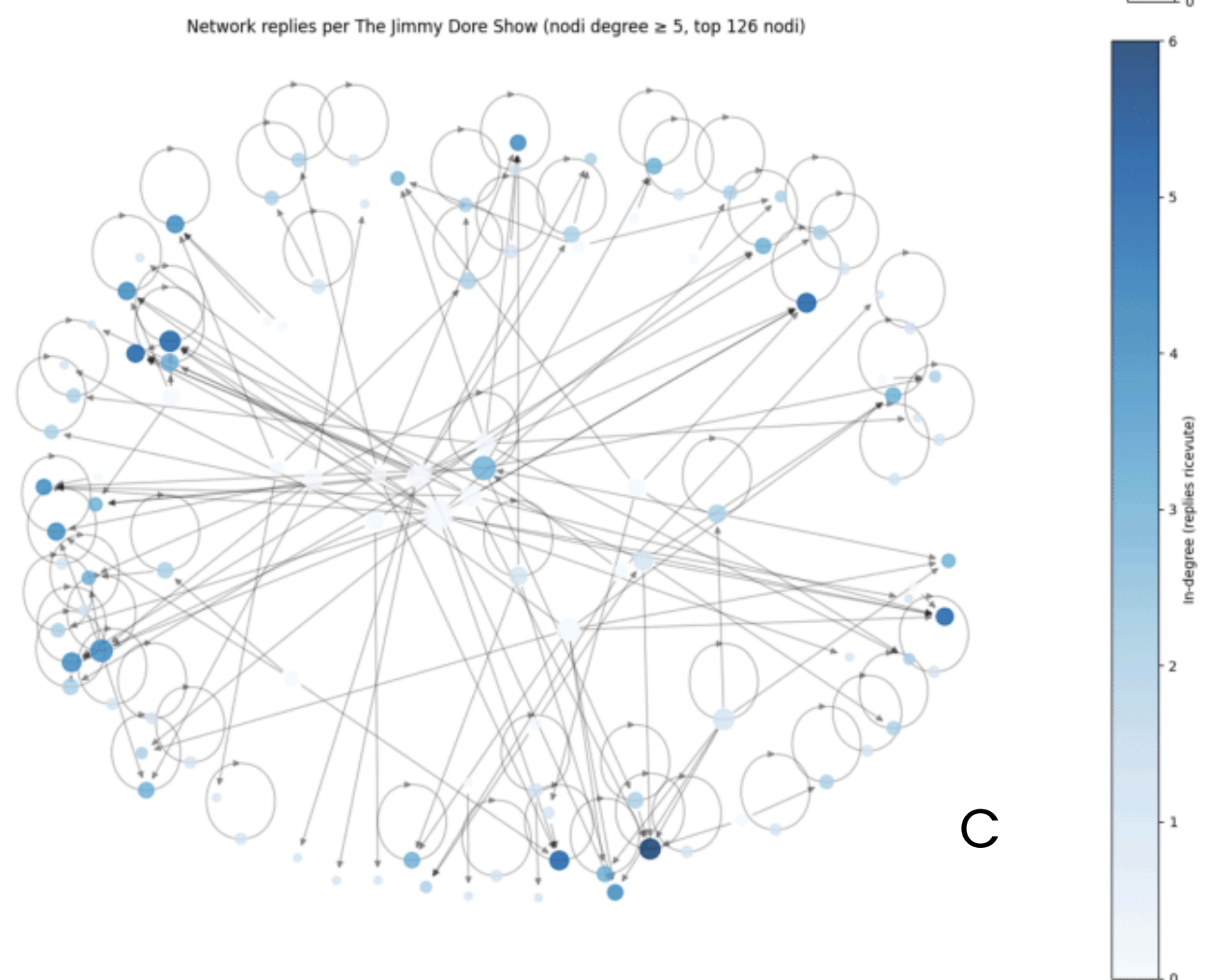
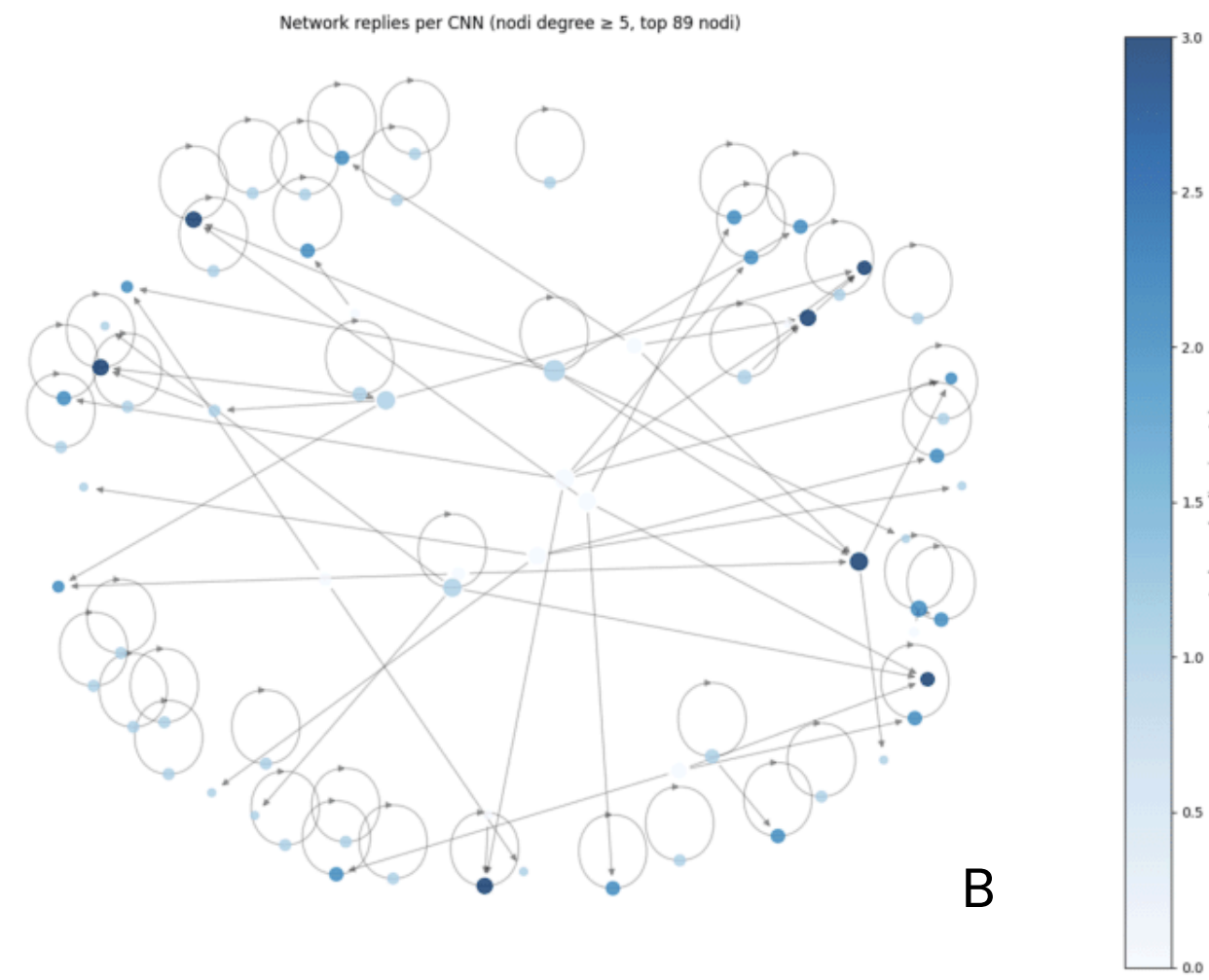
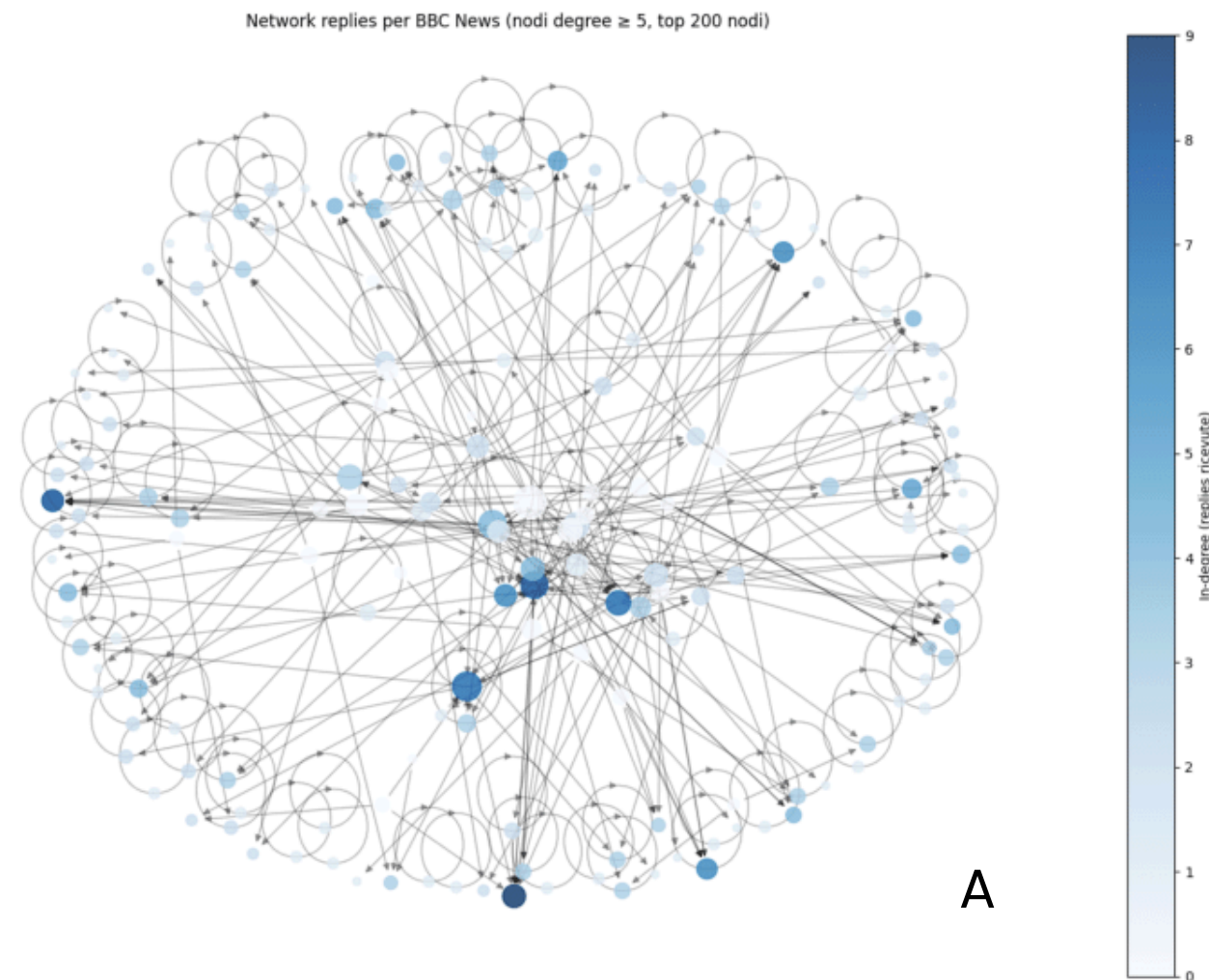
Do the emotions used by channels in their content production translate into similar emotional reactions among users? Are the average emotional profiles of titles and comments aligned within each channel?

- All correlations are weak
- Correlation shows relatively stronger coherence between Title sentiment and Avg Sentiment in the comments for Mainstream channels
- Consequence of: mixed communication strategy; bimodal distribution of sentiment in the comments; loyal community

KEY INSIGHTS OVERVIEW

- A. BBC
- B. CNN
- C. The Jimmy Dore Show
- D. The Young Turks

- Size: Nodes Degree
- Colour Intensity: In-Degree



KEY INSIGHTS OVERVIEW

Metrics computed for the networks

Metric	BBC News	CNN	The Jimmy Dore Show	The Young Turks
Nodes (Number of Users)	2747	1089	1534	1801
Edges (Number of Replies)	3357	1067	1616	1703
Average degree (in+out)	2.44	1.96	2.11	1.89
Density	0.00	0.00	0.00	0.00
Connected components (undirected)	202	185	159	322
Average in-degree centrality (received replies)	1.22	0.98	1.05	0.95
Average out-degree centrality (sent replies)	1.22	0.98	1.05	0.95

KEY INSIGHTS

CONCLUSION

- Individual respond primarily to **thematic framing** rather than sensationalism or scandalistic language
- Main topics in **highest consistent engagement** channels *were* **polarizing** (conspiracies, criticism to political affiliations, etc.)
- With respect to our initial hypothesis and given *our choices* for the analysis we formulate **three new categories**:
 - *Institutional broadcaster* (BBC) with high predictability, broad but centralized interaction, consistent with an informational mandate
 - *Commercial/partisan leaders* (CNN and TYT) with high volume but high extremes, relying on sensational or polarizing events to spike engagement
 - *Niche radical channel* (TJDS) characterized by intense but limited participation

Limitations to the analysis:

- Definition of Alternative (Selection Bias)
- Virality Detection (cap to 200 comments per video)
- Sampling for the network analysis
- Lack of sentiment analysis for replies to detect polarized arguments or communities

A background image showing a hand holding a newspaper, with the text 'THANK YOU' overlaid in a white rounded rectangle.

THANK YOU

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